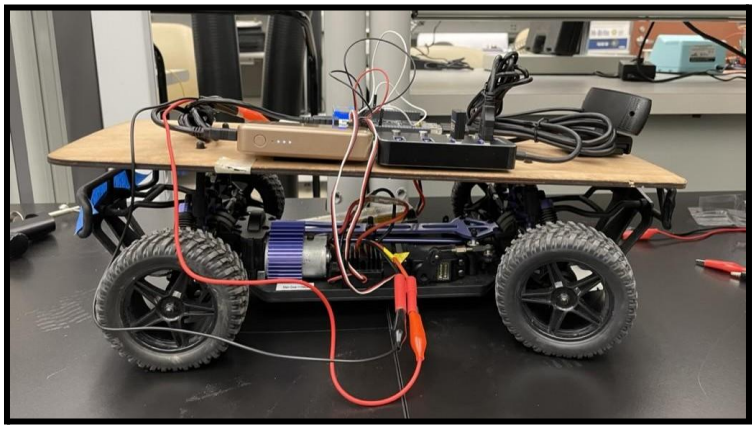
COMP 424 ELEC 424/553 - Final Project: Self-Driving Car

200 points

"Ask any racer. Any real racer. It don't matter if you win by an inch or a mile. Winning's winning." - Dominic Toretto



Overview

Autonomous vehicles are all the rage. Waymo, Rivian, Zoox, Tesla, …. you name it. It's an exciting time, and we don't want to miss out on the action.

You will work as a team to program an autonomous RC/toy car. If it's autonomous, then of course RC (remote control) loses its meaning, but if we say toy car then it might be ambiguous. So, we'll say RC car often just because it implies a small toy car with electronics.

To make this happen, you can use the code from project 3 and previous years as a starting point: link. You can literally start from their scripts, just be sure to cite your sources in your Hackster project description and in any code files. Those scripts take an existing Python script meant for a Raspberry Pi 3 attached to an RC car and modify it to work with the Raspberry Pi 4, Beaglebone Black (BBB), or BeagleBone AI-64 (BBAI64) attached to the RC cars in Ryon. Your job is to make this system work on a Raspberry Pi 5 (RPi 5). You will not only enable the car to perform lane keeping, but also to stop at a red stop box on the ground and (if you are a graduate team) to identify objects using machine learning. Grading will be based primarily on the performance of your car on any of the tracks laid out in PCF3, Ryon B10, or Ryon B12.

NOTE: Even though some prior scripts use the Raspberry Pi 4, the motor control scheme was different to how we are controlling the motors with our Raspberry Pi 5.

Rubric

1. **(120 points) In person or submitted recorded video demonstration (here) of self-driving functionality. In person demonstrations can be done during instructor or TA office hours.**
    a. **(40 points) [30 points if graduate team] Speed encoding**
        i. (20/15 pts) System uses speed encoder to maintain a consistent speed

ii. (20/15 pts) System implements driver to precisely measure timings between optical encodings and these timings are used to adjust the speed control

b. (40 points) [30 points if graduate team] <u>Lane keeping</u>

i. (25/20 pts) Car is able to follow the center of the lane for half of the track

ii. (25/20 pts) Car is able to follow the center of the lane for all of the track

Note: I define **following** as any edge of the car not moving more than 6 inches away from the lane marker closest to the car. Ask us if you are encountering problems with staying this close to the lanes.

c. (40 points) [30 points if graduate team] <u>Red stop paper on ground</u>

i. (30/20 points) Car stops at first stop paper, then resumes lane keeping

ii. (30/20 points) Car stops permanently at second/final stop paper

d. [30 points if graduate team] (not required for other teams)

i. (10 points) Run image recognition successfully using webcam with video output displayed on laptop

ii. (10 points) Estimate frame rate of image recognition

iii. (10 points) Record video output of image recognition

2. (35 points) Hackster page on car development, performance, and PID tuning - include the following elements in a project submitted to this community.

a. (1 point) Name

i. Put the team name here that you created

b. (1 point) Cover image

i. Exercise your creativity on your choice of an image

c. (1 point) Elevator pitch

i. 1 sentence on what your system does

d. (1 point) Components and apps

i. Include components (webcam, RPi 5, optical speed encoder, portable charger) and 1 app (openCV)

e. (1 point) Teammates - please have team members' Hackster accounts linked to the project so that they display on the page

f. (26 points) Story

i. (2 points) 1-3 sentences introducing the project and what the car does. You must cite that this work draws from the following Instructable:

- User raja_961, "Autonomous Lane-Keeping Car Using Raspberry Pi and OpenCV". Instructables. URL: https://www.instructables.com/Autonomous-Lane-Keeping-Car-Using-Raspberry-Pi-and/
- Also be sure to cite any existing 424 projects on that site that you used

ii. (4 points) 3-5 sentences about how you determined the resolution, proportional gain, derivative gain, and (optionally) integral gain to use for the car.

iii. (4 points) 3-5 sentences about how you handled the stop paper and (if graduate team) an additional 3-5 sentences about how you handled object recognition.

iv. (4 points) On one plot show error, steering duty cycle, and speed duty cycle versus frame number for an entire run of the track

v. (4 points) On one plot show error, proportional response, derivative response, and (optionally) integral response versus frame number for the same entire run of the track as used for the previous bullet point.

- The proportional response is the proportional gain times the error
- The derivative response is the derivative gain times the derivative of the error
- (optionally) The integral response is the integral gain times the integral of the error

vi. (4 points) Picture of the vehicle with your hardware on it

vii. (4 points) Video of the vehicle completing the course with your hardware and software running on the vehicle

g. (4 points) Code

i. Please upload any .py files, C files (including driver code for optical encoder), and/or other code used for the project

- You must include your modified device tree source file (.dts)

ii. You must cite the Instructable that the main .py file comes from at the top the file:

- User raja_961, "Autonomous Lane-Keeping Car Using Raspberry Pi and OpenCV". Instructables. URL: https://www.instructables.com/Autonomous-Lane-Keeping-Car-Using-Raspberry-Pi-and/
- Also be sure to cite any other existing 424 projects on Hackster that you used

3. (20 points) Evaluation of teammates' effort

a. (5 points) Fill out form here estimating teammates' effort

b. (15 points) The average of teammates' estimates of your effort

Note: The evaluation will be about whether or not everyone did at least their fair effort.

4. (20 points) Submission of relevant commented source code files to Canvas

a. Code attempts to achieve objectives/requirements stated in instructions

b. Code reasonably includes comments (1 comment at least every 5 lines)

c. The following file(s) must be submitted in their source form (e.g., .py, .c, .dts) - not a PDF. Only 1 team member needs to submit the code.

i. Any .py file(s) and/or other code used for the project
ii. You must include your modified device tree source file (.dts)
iii. We should be able to rerun your python code, don't just give us a file that has functions without the script that calls those functions

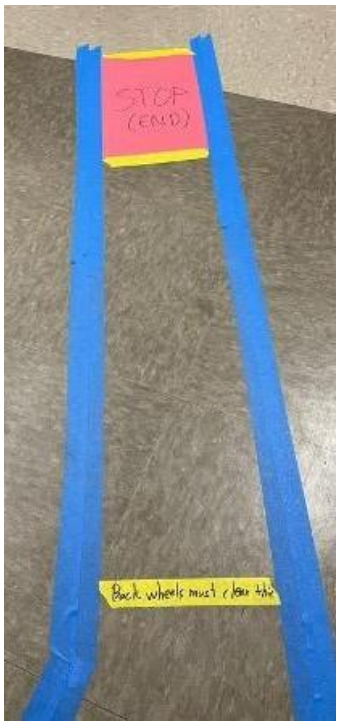
5. **(5 points) Return equipment to your locker in PCF3**
   a. Equipment should include:
      i. RC car (chasis, motor, & ESC), batteries, and charger
      ii. Portable Battery
      iii. Speed Encoder
      iv. Raspberry PI & SD Card
      v. WebCam
      vi. Breadboard
      vii. LCD Display
      viii. USB-C Cable
      ix. Jumpers (provided in a box in PCF3)
   b. Take a picture of the equipment in your locker once it is all there; be sure to submit the picture on Canvas

Guidelines

- **General**
  - Back up your code frequently! Your RPi 5 could get fried.
    - If your microSD card hits issues, you could re-flash a card with my image I gave for project 3 and add your code back.
- **Track**
  - The picture shows one of the tracks. The start is on the left, and the end is on the right. Your car must successfully navigate it, stopping at stop boxes (red paper on the ground). The car shouldn't continue after the 2nd/final stop box.

  - The following picture shows the start where the car must be placed

- The following picture shows the end where the car must stop - specifically, the car must stop after the yellow tape saying "Back wheels must clear this"
    - Note the "stop box" - it's a red piece of paper on the ground :)
    - Similarly, there is a red stop box in the middle of the track as well
- The rest of these guidelines detail requirements and advice around speed control through optical encoding, lane keeping, and the stop boxes



- **Code to start with**
    - Start with one of the 424 project files here and/or the lovely Python file provided on Instructables (Autodesk, Inc.) here by user raja_961 in combination with your project 3 code
        - Note that the Instructables code just turns the car left or right in a binary fashion
    - For our course it is required that you perform "analog" steering, not just full left or full right; This will be done via PID (see later section)
    - I got things working by keeping the speed fixed low and using PD (proportional derivative control) with the steering
    - **You should only run the core loop of the code (the loop that performs lane detection and motor control) for a limited number of times - otherwise the speed motor will keep on going even when you ctl+c the running Python code! This means you also need to add a line after that loop that sets the motor back to neutral so the ESC stops moving the motors.**
    - **As a backup, you can (1) have a Python function that sets motor outputs to neutral and (2) disconnect the 7.2V battery for the car when needed. I've had to pick up the car sometimes and do one of these two since it was still running. Turning the switch of f the ESC should also do the trick.**
- **Lane keeping**
    - Once you've given me your MAC address info and I have asked IT to assign a fixed IP address to your Linux device (IP addresses will appear here), you will generally interact with the car by wirelessly ssh'ing into the Pi 5 attached to it (using the fixed IP address rather than raspberrypi.local).
        - While you're waiting on the fixed IP, you can use the IP displayed on the LCD that you set up in project 3
    - By doing so, you can wirelessly tell the car to start its Python lane keeping/self driving script. If you ssh in with X forwarding as I mentioned in class, you will

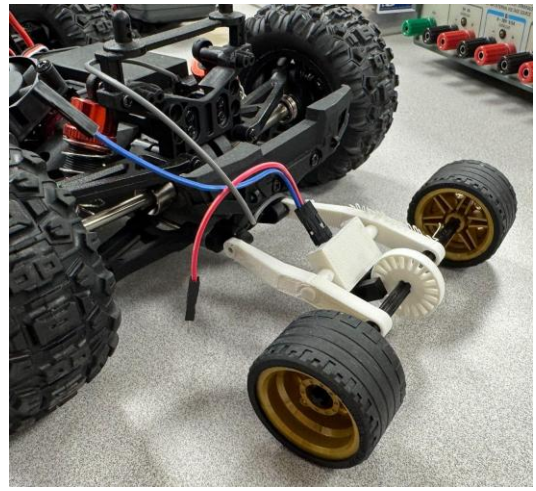
also get a computer vision stream to your laptop. I recommend commenting out all of the image showing functions in the script except for the one that shows the lane detection and intended path.

- MobaXterm on Windows will do X forwarding by default when you SSH into the Pi
- On Mac, you should SSH with the -Y flag after installing XQuartz
  - E.g., **ssh -Y pi@192.168.1.1**
    - But you would replace the IP address with your Pi's

- You will need to resize (downsize) the image in order for processing to occur at any reasonable rate
- This is one of your parameters - it will have to be fairly low, even 720p is often too high (although the Pi 5 may be able to handle this more; you can explore)
  - You can get away with a pretty low resolution here, try out different stuff
  - You can set the webcam to run at a lower resolution using opencv, the following command shows you what resolutions your webcam supports:
    - v4l2-ctl -d /dev/video2 --list-formats-ext
  - Then you can use opencv to resize it even smaller
- Images/video window not showing? You may need to add `cv2.waitKey(1)` in your core loop if it is not there already
  - More info - MLK, URL: https://machinelearningknowledge.ai/opencv-cv2-waitkey-tutorial-with-examples/

- **PID/PD controller for steering**
  - The project programs on Hackster implement a PD controller for steering
  - You may need to change the proportional and derivative gains
  - You have to think about how the error gets mapped by the PD controller to action (steering)
  - Once aspect of this is the range of values that will be output by the PD controller
    - The PD controller should really only cause the steering PWM to go between 6% and 9% (it would produce a PWM of 7.5% for zero error)
  - This is why I put a plotting requirement in the rubric for the report - it is helpful to plot out different aspects of the error/PD/steering and see how they are interacting
  - You can start out by setting P gain to some low value and D gain to zero
    - As you increase the P gain, the car should be more responsive to error, but eventually it will oscillate around the desired trajectory
    - Increasing the D gain allows you to reduce this oscillation
  - By plotting and/or testing, you should be able to find values that get you there

- I highly encourage you to carefully observe the car's behavior and sometimes plot things to figure out how to work with P and D, rather than just guessing values (trust me, I've done the latter a lot…)
- **Encoder-based controller for speed**
  - You have been given an optical encoder like this one, put inside a 3D printed white mount (see the picture)



  - This mount can be attached to the back or front of your car, enabling the encoder wheel to spin on the ground as your car moves.
    - You can be as creative as needed to attach the mount, e.g., you can even drill holes in the car chassis if you want
    - In some cases, you can use two screws to secure the encoder mount to the car
      - We have provided a bag of screws that you can grab two screws from in PCF3
        - You can find them at the end of the left benchtop when you come into the room
        - The bag is on top of a manila envelope with 424 written on it
  - The optical encoder will output high or low depending on whether or not light can shine through the wheel
    - Note - the encoder has 3 pins:
      - 5V/VCC - connect this to 5V your Pi
      - GND - connect this to ground of your Pi
      - OUT - connect this to a GPIO pin of your Pi
  - You should modify the project 2 driver to count the time between optical encoder activations, since the optical encoder will act like a button
    - You are required to use gpiod rather than the older deprecated method
  - This requires modifying the device tree of the RPi 5 to set up a pin as the input reading the optical encoder, just like you did for project 2

    - Your driver should be able to recognize changes in the optical encoding as button presses, and you can add extra code to your driver to calculate the timings
        - Reject timings below some number of ms based on your testing, those are issues in the encoder like switch debouncing
        - You can also grab capacitors from the shelves in PCF3 to put between ground and the OUT output of the encoder, since this will help "debounce"; Try different capacitance values.
    - Somewhere in your main python script that is handling the system, you will want to periodically find out the rate of encoder wheel turning that your button/encoder driver is recording
        - You could make this a module parameter and read it in sysfs, then adjust motor control accordingly to maintain constant speed, or you could do file operations to where the read function outputs the rate
- **"Stop boxes"**
    - Again, the two stop boxes are red paper on the ground
    - Brainstorm/do research/look at prior teams' code to find a good solution that involves processing of the image to see the red area and accordingly stop the vehicle
        - If at the first stop box, continue moving after stopping
        - If at the second/final stop sign (end of the track), stop permanently
    - You will have to be conscious of how this check/processing affects the performance of your vehicle in terms of lanekeeping
        - Feel free to employ tricks here, e.g. only check for stop boxes every 3rd frame
- **Folks enrolled in 553 only: Objection Recognition**
    - If you are enrolled in the 553 version of the course, then this is your extra requirement
    - You must install and run some sort of object recognition algorithm on the Pi that applies bounding boxes and labels, e.g., YOLOv5
        - Use a resolution of 320 by 320
    - Estimate and record the frame rate of YOLOv5
    - Include a video of the whole setup, showing your Pi with the webcam and the frames being streamed to your laptop with bounding boxes and labels, also showing how the frame changes when you move the webcam
        - The video should be included on your Hackster page (no need to submit it to Box)